

Renjie Ma

PhD, Associate Professor

State Key Laboratory of Robotics and Systems
Department of Mechatronic Control and Automation
Harbin Institute of Technology

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Academic Appointments

Associate Professor, Mechanical Engineering Jan 2026-Present
Harbin Institute of Technology, Harbin 150001, China

Postdoctoral Researcher, Mechanical Engineering Jul 2022-Dec 2025
Harbin Institute of Technology, Harbin 150001, China

Visiting Scholar, Electrical and Mechanical Engineering Jul 2026-Aug 2026
Adelaide University, Adelaide, SA 5005, Australia

Visiting Scholar, Electrical and Mechanical Engineering Jul 2025-Aug 2025
Adelaide University, Adelaide, SA 5005, Australia

Education Backgrounds

Doctor of Philosophy (PhD), Control Science and Engineering July 4, 2022
Harbin Institute of Technology, Harbin 150001, China
Advisors: Prof. Peng Shi (FATSE, MAE, FIEEE, FIET, FIEAust, FCAA) and Prof. Ligang Wu (FIEEE, FCAA)
Thesis: Attack-Resilient Control and Secure Estimation for Cyber-Physical Systems

Bachelor of Engineering, Automation July 1, 2016
Hefei University of Technology, Hefei 230009, China
Advisor: Dr. Tao Zheng
Thesis: Adaptive Control for Direct Current Motor-Actuated Unmanned Ground Vehicle

Research Interests

- **AI-Based Control Theory and Application:** Modeling theory and learning control of NFLs, Safe learning motion planning and control, Verifiable continual Learning.
- **Optimization and Control for Complex Systems:** Data-driven control; Model predictive control; Adaptive control; Aperiodic-sampled control.
- **Robotic Control:** Biomimetic soft robots; Robotic manipulator; Mobile robots.

Experimental Studies

- Event-triggered setpoint tracking control of wheeled vehicles via descriptor reduced-order observer
(<https://www.youtube.com/watch?v=4aqSejbu4HE>, <https://www.youtube.com/watch?v=fW3E4FxjJAY>)
Article: IEEE Transactions on Intelligent Transportation Systems, vol. 26, no. 7, pp. 10308-10319, 2025.
- Memory-event-triggered predictive control for trajectory tracking of wheeled vehicles
(<https://www.youtube.com/watch?v=eqZ0xHaa5kA>)
Article: IEEE Transactions on Intelligent Vehicles, vol. 11, no. 4, pp. 500-510, 2026.
- Event-triggered robust backstepping tracking control of wheeled vehicles via adaptive state observer
(<https://www.youtube.com/watch?v=tMyuvCNPxb4>)

Article: IEEE Transactions on Industrial Electronics, Under revision.

Coordinated Scientific Grants

Imitation Learning Control and Robustness Assessment for Soft Robot Toward Space Exploration

National Natural Science Foundation of China, Young Scientists Fund, No. 62503133, Coordinator,
Jan 2026-Dec 2028, 300,000 RMB.

Fully-Actuated System Theory-Based Imitation Learning Control and Application on Soft Robots

National Natural Science Foundation of China, Basic Science Center Project, No. 62188101, Co-Coordinator.
Jan 2026-Dec 2026, 200,000/70,056,000 RMB.

Imitation Learning Control for Soft Robots

Natural Science Foundation of Heilongjiang Province, General Program (GP), No. LH2024F026, Coordinator.
Dec 2024-Dec 2027, 100,000 RMB.

Data-Driven Safe Tracking Control and Robust Assessment for Soft Robots

Natural Science Foundation of Chongqing Municipality, GP, No. CSTB2023NSCQ-MSX0625, Coordinator.
Jul 2023-Jul 2026, 100, 000 RMB.

Event-Triggered Safe Learning Control for Soft Robots

China Postdoctoral Science Foundation, GP, No. 2023M730864, Coordinator.
Jul 2023-Jul 2025, 80, 000 RMB.

Safe-Learning Robotic Motion Planning and Control

Basic Scientific Research Business Expenses of Central Universities, No. HIT.NSFJG202445, Coordinator.
Jul 2025-Nov 2025, 36, 000 RMB.

Data-Driven Safe Motion Control for Electromagnetic-Actuated Microrobot

Heilongjiang Provincial Postdoctoral Science Foundation, GP, No. LBH-Z22150, Coordinator.
Nov 2022-Nov 2024, 100, 000 RMB.

Self-Evolutionary Decision Planning of Intelligent Vehicles Based on Trusted Cognition in an Open Environment

Open Project of MoE Key Laboratory of Cognitive Intelligence and Content Security, RZZN202403, Coordinator.
Jan 2024-Dec 2024, 70, 000 RMB.

Safety Assessment and Attack Defense Control of Industrial Robots for Production Operation

Harbin Institute of Technology, Youth Talent Cultivation Program, Coordinator.
Sep 2024-Sep 2027, 150, 000 RMB.

Data-Driven Motion Planning and Control for Microrobots

Harbin Institute of Technology, Start-Up Fund, Coordinator.
Jul 2022-Feb 2025, 100, 000 RMB.

Academic Publications

Journal Papers:

Z. Hu and R. Ma*, Event-triggered robust tracking control of a networked commercial aircraft system during ascent under communication constraints, *IEEE Transactions on Network Science and Engineering*, DOI:10.1109/TNSE.2026.3729974.

Z. Hu, Q. Liu, Z. Gao, Z. Wang, R. Zhang, and R. Ma*, Self-triggered fault-tolerant control of cyber-physical Markovian jump systems via descriptor sliding mode observer, *IEEE Transactions on Industrial Cyber-Physical Systems*,

vol. 4, pp. 161-170, 2026.

Z. Hu[†], **R. Ma**^{†*}, S. Zhang, H. Wang, X. Liu, and D. Zhao, Memory-event-triggered predictive control for trajectory tracking of autonomous ground vehicle, *IEEE Transactions on Intelligent Vehicles*, vol. 11, no. 4, pp. 500-510, 2026

Z. Hu[†] and **R. Ma**^{†*}, Adaptive event-triggered tracking control via switching functions. *Automatica*, vol. 184, num. 112813, 2026.

R. Ma, Z. Hu, R. Yang, and L. Wu*, Aperiodic-sampled neural network controllers with closed-loop stability verifications, *Automatica*. vol. 62, num. 112573, 2026

R. Ma^{†*}, Y. Yu[†], S. Zhang, and P. Shi*, Aperiodic-sampled fault-tolerant control for autonomous ground vehicle via descriptor reduced-order observer, *IEEE Transactions on Intelligent Transportation Systems*, vol. 26, no. 7, pp. 10308-10319, 2025.

Z. Hu, R. Su, K. Zhang, R. Wang, and **R. Ma**^{*}, Resilient frequency estimation for renewable power generation against phasor measurement unit and communication link failures, *IEEE Transactions on Circuits and Systems II: Express Briefs*, vol. 72, no. 1, pp. 233-237, 2025.

Z. Hu, R. Su, V. Veerasamy, L. Huang, and **R. Ma**^{*}, Resilient frequency regulation for microgrids under phasor measurement unit faults and communication intermittency, *IEEE Transactions on Industrial Informatics*, vol. 21, no. 2, pp. 1941-1949, 2025.

Z. Hu, H. Qiu, H. H. Alhelou, X. Xie, and **R. Ma**^{*}, Resilient distributed load frequency control for interconnected power systems with PEVs and wind turbines against temporary PMU faults, *IEEE Internet of Things Journal*, vol. 11, no. 23, pp. 38719-38727, 2024.

Z. Hu[†], **R. Ma**^{†*}, B. Wang, Y. Huang*, and R. Su, A general resiliency enhancement framework for load frequency control of interconnected power systems considering internet of things faults, *IEEE Transactions on Industrial Informatics*, vol. 20, no. 9, pp. 10695-10705, 2024.

R. Ma^{*} and Z. Hu*, Deep neural network-controlled safety-critical systems with uncertainty resilience, *IEEE Internet of Things Journal*, vol. 11, no. 13, pp. 24061-24072, 2024.

R. Ma, Z. Hu, L. Xu, and L. Wu*, Distributed secure estimation against sparse false data injection attacks, *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, vol. 54, no. 5, pp. 2685-2697, 2024

Z. Hu, R. Su, K.-V. Ling, Y. Guo, and **R. Ma**^{*}, Resilient event-triggered MPC for load frequency regulation with wind turbines under false data injection attacks, *IEEE Transactions on Automation Science and Engineering*, vol. 21, no. 4, pp. 7073-7083, 2024.

R. Ma[†], Z. Hu[†], H. Yang, Y. Jiang, M. Huo, H. Luo*, and R. Yang*, Adversarial FDI attack monitoring: Toward secure defense of industrial electronics, *IEEE Industrial Electronics Magazine*, vol. 18, no. 2, pp. 48-57, 2024.

H. Yang and **R. Ma**^{*}, Sparse attack reconstruction for cyber-physical systems via descriptor reduced-order observer, *Nonlinear Analysis: Hybrid Systems*, vol. 50, num. 101380, 2023.

Z. Hu, R. Su, K. Zhang, Z. Xu, **R. Ma**^{*}, Resilient event-triggered model predictive control for adaptive cruise control under sensor attacks, *IEEE/CAA Journal of Automatica Sinica*, vol. 10, no. 3, pp. 798-800, 2023.

R. Ma^{*} and P. Shi, Secure state estimation for cyber-physical systems under sparse data injection attacks: A switched counteraction approach, *International Journal of Control*, vol. 95, no. 4, pp. 963-974, 2022.

R. Ma^{*}, P. Shi, and L. Wu, Sparse false injection attacks reconstruction via descriptor sliding mode observers, *IEEE*

Transactions on Automatic Control, vol. 66, no. 11, pp. 5369-5376, 2021.

R. Ma*, P. Shi, and L. Wu, Active resilient control for two-dimensional systems under denial-of-service attacks, *International Journal of Robust and Nonlinear Control*, vol. 31, no. 3, pp. 759-771, 2021.

R. Ma*, P. Shi, and L. Wu, Dissipativity-based sliding mode control of cyber-physical systems under denial-of-service attacks, *IEEE Transactions on Cybernetics*, vo. 51, no. 5, pp. 2306-2318, 2021.

R. Ma*, P. Shi, Z. Wang, and L. Wu, Resilient filtering for cyber-physical systems under denial-of-service attacks, *International Journal of Robust and Nonlinear Control*, vol. 30, no. 5, pp. 1754-1769, 2020.

R. Ma, X. Shao, J. Liu, and L. Wu*, Event-triggered sliding mode control of markovian jump systems against input saturation, *Systems & Control Letters*, vol. 134, num. 104525, 2019.

R. Yang*, R. Yang, **R. Ma**, and W. X. Zheng, Data-driven control of discrete-time two dimensional Fornasini-Marchesini systems with disturbances. *International Journal of Robust and Nonlinear Control*. DOI: 10.1002/rnc.70685.

Y. Yan, C. Zhang, D. Han, **R. Ma**, H. Wang, D. Pi, and G. Yin, How do lane-keeping autonomous vehicles avoid being bullied by unreasonably cut-in human-driven vehicles? *Control Engineering Practice*, vol. 168, num. 106668, 2026.

Y. Yan, C. Du, W. Wang, **R. Ma**, Y. Wang, H. Wang, D. Pi, and Y. H. Chen, Road surface status recognition and estimation: State-of-the-art review and research perspectives, *Chinese Journal of Mechanical Engineering*, vol. 39, num. 100370, 2026.

Y. Tian, **R. Ma**, W. Luo, Y. Gao, and L. Wu*, Secure control for remote networked stochastic systems via integral sliding mode, *ISA Transactions*, vol. 146, pp. 208-220, 2024.

Y. Jiang*, S. Wu, **R. Ma**, M. Liu, H. Luo*, and O. Kaynak, Monitoring and defense of industrial cyber-physical systems under typical attacks: From a system and control perspectives, *IEEE Transactions on Industrial Cyber-Physical Systems*, vol. 1, pp. 192-207, 2023.

Journal Preprints:

R. Ma*, S. Zhang, W. Liu, Z. Hu, and P. Shi, Learning robust controller for linear parameter-varying systems from data. Preprint to *IEEE Transactions on Automatic Control* (arXiv: 2506.08366). Submission Date: Jun 09, 2025. Revision Under Review.

R. Ma, J. Xiong, W. Liu*, and Z. Hu, Data-driven event-triggered predictive control of linear parameter varying systems. Preprint to *Automatica*. Submission Date: May 05, 2026. Under Review.

R. Ma*, Z. Hu, W. Liu, and C. Huang, Event-triggered robust backstepping tracking control of wheeled vehicles via adaptive state observer. Preprint to *IEEE Transactions on Industrial Electronics*. Submission Date: June 02, 2026. Revision.

R. Ma*, Learning fault-tolerant controller via neural descriptor observer, Preprint to *Automatica*. Submission Date: Jun 05, 2026. Under Review.

Z. Hu and **R. Ma***, Disturbance Observer-Based Finite-Time Tracking Control for Consumer Soft Robotic Systems, Preprint to *IEEE Transactions on Aerospace and Electronic Systems*. Submission Date: Jun 26, 2026. Under Review.

S. Zhang, Z. Hu, and **R. Ma***, Reinforcement learning-based safe tracking control of unmanned ground vehicles. Preprint to *IEEE Transactions on Vehicular Technology*. Submission Date: May 24, 2026. Under Review.

H. Zheng, M. Shi, Z. Hu, and **R. Ma***, Adaptive recurrent neural network control for safety-critical systems under

DoS attacks. Preprint to *IEEE Transactions on Control of Network Systems*. Submission Date: Aug 30, 2026. Under Review.

R. Ma and P. Shi*, Adaptive safety-critical output feedback tracking control via residual neural network. Preprint to *IEEE Transactions on Automatic Control*, Submit Soon.

Conference Papers:

J. Xiong, S. Zhang, Y. Li, Z. Hu, B. Yan, and **R. Ma***, Robust event-triggered trajectory tracking control of robotic manipulator via sliding mode, in *2026 International Conference on Artificial Intelligence and Data Intelligent Processing (AIDIP 2026) & 2026 International Conference on Deep Learning and Automation (ICDLA 2026)*, Accepted.

R. Zhang, Z. Hu, and **R. Ma***, Adaptive event-triggered robust trajectory tracking control of soft robots, in *24th IEEE International Conference on Industrial Informatics (INDIN 2026)*, Accepted.

Y. Wu, Z. Qu, X. Chang, Z. Hu, R. Yang, and **R. Ma***, Robust tracking for electromagnetic microbot via sliding mode control, in *49th Annual Conference of IEEE Industrial Electronics Society (IECON-2023)*, DOI: 10.1109/IECON51785.2023.10311964.

R. Ma, J. Liu, and L. Wu*, Privacy-enabled secure control of fog computing aided cyber-physical systems, in *45th Annual Conference of IEEE Industrial Electronics Society (IECON-2019)*, pp. 509-514, 2019.

Z. Hu, C. Wang, H. Ye, M. Wang, Z. Xu, and **R. Ma**, Resilient control of multi-energy power generation in smart cities, in *the 23rd IFAC World Congress (IFAC 2026)*. Accepted.

R. Yang, R. Yang*, **R. Ma**, Y. Zhu, and P. Shi, Data-based stabilization of unknown discrete-time 2-D Roesser systems, in *International Conference on Machine Learning and Cybernetics (ICMLC-2024)*, pp. 559-564, 2024.

Patents

S. Zhang, **R. Ma**, Z. Hu, W. Liu, Z. Wang, R. Zhang, and J. Xiong, Aperiodic-sampled deep neural network controller synthesis based on closed-loop stability verification, China, 202511395484.9.

R. Ma, J. Xiong, Z. Hu, Z. Wang, and R. Zhang, Event-triggered longitudinal motion control of commercial aircraft in the climbing process via sliding mode, China, 202511235814.8.

J. Lu, X. Wang, Y. Zhou, L. Li, L. Yao, **R. Ma**, and J. Qiao, A highly sensitive flexible vibration sensor based on electret, China, 202511233724.5.

S. Zhang, **R. Ma**, Z. Hu, Z. Wang, W. Liu, R. Zhang, and X. Chang, An event-triggered data-driven robust tracking control of LPV systems, China, 202511235814.8.

R. Ma, Y. Feng, Z. Wang, Z. Hu, X. Chang, and D. Zhou, An event-triggered model predictive control of soft robots, China, 202511237704.5.

R. Ma, Z. Qu, Z. Hu, S. Zhang, Z. Wang, J. Qiao, and D. Zhou, A finite-time robust trajectory tracking control of soft robots, China, 202510327984.2.

R. Ma, Z. Hu, Y. Gao, J. Qiao, and J. Lu, Stability assessment and uncertainty bound estimation for a class of nonlinear control systems, China, 202310801095.6.

J. Qiao, **R. Ma**, D. Fu, B. Zhou, and H. Shi, A method for determining the location and type of internal defects in three-dimensional packaging of multilayer flip-chip, China, 202410227091.6.

H. Wang, H. Luo, **R. Ma**, X. Qiao, M. Huo, S. Wu, and Z. Ning, A distributed fault diagnosis method for interconnected systems with robust performance in the communication links, China, 202410400894.7.

L. Yao, C. Jin, X. Jin, J. Qiao, **R. Ma**, and J. Lu, Multi-degree-of-freedom floating rotor self-balancing wave power generation device, China, 202311773931.0.

Academic Presentations

Oral Presentation in ICICIC 2026, Stability-Certified Aperiodic-Sampled DNN Controllers, Nov 22, 2026, Chongqing, China.

Oral Presentation in INDIN 2026, Adaptive Event-Triggered Robust Trajectory Tracking Control of Soft Robots, Jul 28, 2026, Melbourne, Australia.

Oral Presentation in ICICIC 2025, Learning Controllers for LPV Systems From Noisy Data, Aug 18, 2025, Kitakyushu, Japan.

Invited Talk in Adelaide Control Systems Seminar, Learning Robust Control Strategies From Expert Demonstrations, Aug 06, 2025, Building IW 5.55, Engineering North, Adelaide University, Australia.

Oral Presentation in ICICIC 2024, Robust Motion Tracking Control for Soft Robots with Uncertainty Resilience, Sep 20, 2024, Qinhuangdao, China.

Oral Presentation in IECON 2023, Robust Tracking for Electromagnetic Microrobot via Sliding Mode, Oct 18, 2023, Singapore.

Oral Presentation in IECON 2019, Privacy-Enabled Secure Control of Fog Computing-Aided CPS, Oct 16, 2019, Lisbon, Portugal.

Selected Honors/Awards

2026 Best Paper, International Conference on Deep Learning and Automation (ICDLA 2026).

2022 Outstanding PhD Award, Harbin Institute of Technology

2021 Chun-Yan Young Talent Program, Harbin Institute of Technology

2020 National Scholarship for Doctor Candidates (No. BSY202006514), Ministry of Education, China

2020 Outstanding Student Award, Harbin Institute of Technology

2020 China Scholarship Council Fellowship (No. 202006120091)

2019 International Academic Exchange Fund (No. 2019276), Harbin Institute of Technology

2017 First-Class Scholarship for Postgraduate, Harbin Institute of Technology

Technical Activities

· **Referee**, National Natural Science Foundation of China.

· **Referee**, Chinese Ministry of Education's Degree Center.

· **Member**, Technical Committee on Embodied Artificial Intelligence of Chinese Association of Automation.

- **Member**, Technical Committee on Data-Driven Control and Monitoring of IEEE Industrial Electronics Society.
- **Member**, IEEE, IEEE Industrial Electronics Society, IEEE Systems, Man, and Cybernetics: Systems Society.
- **Member**, Chinese Association of Automation.

· **Conference Organizing Committee Member:**

3rd International Conference on Intelligent Control and Computing (IC&C2025), Changchun, China
1st CAA Conference on Embodied Artificial Intelligence (CEAI 2024), Shanghai, China

· **Conference Committee Member:**

21st IEEE International Conference on Industrial Informatics (INDIN2023), Lemgo, Germany;
32nd International Symposium on Industrial Electronics (ISIE2023), Helsinki-Espoo, Finland.

· **Conference Special Session Chair:**

2nd Annual Online Conference of the IEEE Industrial Electronics Society, Zoom;
49th Annual Conference of the IEEE Industrial Electronics Society, Singapore;
6th IEEE International Conference on Industrial Cyber-Physical Systems, Wuhan, China;
45th Annual Conference of IEEE Industrial Electronics Society, Lisbon, Portugal.

· **Reviewer for International Journals:**

IEEE Transactions on Automatic Control;
Automatica;
IEEE Transactions on Control Systems Technology;
IEEE Transactions on Cybernetics;
IEEE Transactions on Systems, Man, and Cybernetics: Systems;
IEEE Transactions on Circuits and Systems I: Regular Papers;
ISA Transactions;
International Journal of Robust and Nonlinear Control, etc.

· **Reviewer for International Conferences:**

IEEE Conference on Decision and Control (CDC);
American Control Conference (ACC);
Annual Conference of IEEE Industrial Electronics Society (IECON), etc.

- **Guest Editor**, *Energies* (Special Issue: Hybrid modeling, stability analysis, and control design of future microgrids with distributed renewable energy sources integration)

Teaching Courses

22ME32532: Artificial Intelligence and Optimization Algorithms, Undergraduate course
22ME32524: Intelligent Control for Autonomous Unmanned Vehicles, Undergraduate course

Academic Collaboration

- School of Electrical and Mechanical Engineering, Adelaide University, Australia
- Department of Aeronautical and Aviation Engineering, The Hong Kong Polytechnic University, China
- School of Electrical and Electronic Engineering, Nanyang Technological University, Singapore